Questionnaire Metrics & Data Mapping Guideline

Technical reference mapping questionnaire items (Q1–Q27) to data structures, 14-dimensional feature coordinates, and dynamic dashboard components.

1. AI Data-Driven Cluster Discovery Framework

The analytics portal moves beyond arbitrary tenure-based or demographic segmentation. Agent personas are discovered through multi-dimensional behavioural vectors reflecting digital tool usage, workarounds, satisfaction, and operational dependencies.

> K-means groups the respondents who remain after DBSCAN outlier filtering. The analyst chooses K (3–8, default 5) in the dashboard; outliers are shown as a separate group. Silhouette-based K suggestion is not yet implemented.

Dashboard Analysis Logic:
Each respondent is mapped to a **14-dimensional coordinate vector** spanning 5 core skills (Q7), 6 task-satisfaction ratings (Q27 Tasks 1–6), and 3 behavioural scores (WP, AR, PR). **System Utility Perception (SUP)** is calculated from Q10/Q11 and displayed on the dashboard for diagnostic overview, but is **not** an input to the clustering vector. Discovered cluster centroids are computed dynamically, allowing researchers to explore demographic cross-tabulations without demographic bias influencing the clustering algorithm.

2. AI Classification Vector Definitions

Metric	Code	Data Sources	Formula & Operational Logic
System Utility Perception	`SUP`	Q10, Q11	Average of general digital tool efficiency impact (Q10) and system impact ratings (Q11). Displayed on persona cards; excluded from clustering vector.
Workaround Propensity	`WP`	Q13, Q14, Q23	`min(5.0, 1.0 + reasons × 0.5 + methods × 0.3 + tools × 0.4)`. Uses non-space delimiters (`/[;; ,,]+/`) to parse reasons and tools without splitting compound phrases.
Abandonment Rate	`AR`	Task 1–6 Pain Points	`1.0 + (matched_dropout_tasks / 6.0) × 4.0`. Keyword/semantic detection of process dropouts, password resets, rejected files, forced re-entries, and manual fallbacks.
Peer Reliance	`PR`	Task 1–6 Tools/People, Q23	`min(5.0, 1.0 + peer_signals × 0.5)`. Counts assistance from colleagues, administrators, mentors, and supervisors across journey stages.

Documented Dropout Keywords for AR

- **English Callset**: `"by hand"`, `"converting pdf"`, `"re-entering"`, `"re-enter"`, `"twice"`, `"rejected"`, `"re-upload"`, `"re-uploading"`, `"password reset"`, `"chasing a colleague"`, `"logging back in"`, `"system error"`, `"give up"`, `"giving up"`, `"abandon"`, `"manual"`

- **Traditional Chinese Callset**: `"手動"`, `"手寫"`, `"重複輸入"`, `"重填"`, `"退件"`, `"重新上傳"`, `"重設密碼"`, `"密碼"`, `"重新登入"`, `"強制登出"`, `"被登出"`, `"系統錯誤"`, `"錯誤"`, `"放棄"`

3. Dashboard Field Definitions & Data Mapping

Dashboard Component	Questionnaire Source	Data Processing & UI Mapping
Agent Name	User Input Box	Editable text field for persona naming (e.g. *Digital Trailblazer*). Not derived from survey row.
Position / Title	Q4	Modal rank/title across respondents in the cluster (e.g. *Senior Business Representative*).
Sales Journey Focus	Q26	Categorised distribution percentage of operational focus (Direct Sales vs. Customer Servicing).
Self-Image Style Tags	Q17, Q18, Q19	Dominant workstyle, relationship, and technology self-perception tags for the cluster.
Device / System Familiarity	Q8	Average reported familiarity with company tools (percentage score mapped to progress indicator).
Daily Tasks Breakdown	Q20	Aggregated breakdown of agent daily time distribution across sales, servicing, admin, and prospecting.
Barriers & Contradictions	Q21, Q22	Key qualitative frustrations and operational obstacles reported by cluster members.
Journey Steps & Emotional Trajectory	Q27 (Tasks I–VI)	Six-stage timeline from Task 1 (Scheduling) to Task 6 (Training). Emojis and satisfaction levels map to 1–5 scale. *Note: The CSV has one rating per task* (e.g. `Task1_Scheduling_rating` or `Journey_1_Rating`), rather than separate sub-action letter scores.
Cluster Assignment & Outliers	Internal Classifier	Stored in `agent._dynamicClusterId` (0 to K-1 for standard clusters, and `-1` for DBSCAN outliers). Cluster tab keys map to `archetype` (`c0`, `c1` ... or `outlier`).

> **Data Export & Internal Identifiers**: The active cluster assignment for each respondent is recorded in `agent._dynamicClusterId`. Outlier respondents identified by DBSCAN are assigned `_dynamicClusterId = -1` and are grouped under a dedicated Outlier card in the dashboard.